

From salience to semantics: multilevel hierarchical contingencies organise parent-infant joint attention

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This **valuable** study investigates how perceptual and semantic features of maternal behavior adapt to infants' attention during naturalistic play, providing new insights into the bidirectional and hierarchical organization of early social interaction. The methodology is innovative and overall **solid**, supported by comprehensive multimodal analyses and advanced information-theoretic methods, though some developmental claims warrant further tests of directionality and age effects. The work will be of interest to psychologists, cognitive scientists, and developmental researchers studying early communication, social learning, and methodological innovation in quantifying naturalistic behavior.

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Abstract

Despite strong evidence that children learn more effectively from face-to-face interactions than from screens, we still understand relatively little about the dynamic, adaptive processes through which inter-personal contingency enhances attention and learning during live interactions. In this study, we investigate how social signals during early interactions operate across multiple hierarchical levels, ranging from low-level salience cues to higher-order features. Specifically, we examine how mothers dynamically and reciprocally adjust their behaviours across these levels in response to their infants' attention during play. To achieve this, we developed a suite of novel, information-theory-based methods to quantify naturalistic audio-visual-semantic behaviours. Using time-series analyses, we assessed moment-by-moment associations between infant attention and both lower-order features (e.g., spectral flux of ambient noise and maternal vocalizations, maternal face and hand movement) and higher-order features (e.g., speech information rate, facial expression novelty, semantic surprisal, and toy naming) in tabletop interactions involving 67 mother-infant dyads (5- and 15-month-olds). Our findings suggest that, from early infancy, the information infants perceive is continuously and dynamically modulated across multiple hierarchical levels, contingent on their behaviour and attention. When infants focus on

objects, mothers reduce low-level sensory input, minimising distractions. Conversely, increases in object naming and high-level information content associate with increases in sustained attention. These results indicate that maternal behaviours are both driven by and predictive of infant attention, and that, even from early development, attention involves interactive processes which unfold across multiple levels, from salience to semantics.

1. Introduction

Infants construct and refine predictive models of their world, seeking out stimuli that maximize their learning potential (Gerken et al., 2011 [↗](#); Kidd et al., 2014 [↗](#); Poli et al., 2020 [↗](#)). These predictions allow them to navigate their environments more effectively (Köster et al., 2020 [↗](#)), orienting their attention toward information that optimizes their ability to learn (Berger & Posner, 2022 [↗](#)). Maximizing learning from the environment requires infants to build hierarchical predictive models that span a broad range of representations, from low-level features such as fast-varying fluctuations in amplitude, pitch and luminance, through to higher-order features which operate over slower time-scales, and which arise from the hierarchical integration of low-level features into cognitively meaningful information (Heilbron et al., 2022 [↗](#); Kothinti & Elhilali, 2023 [↗](#)).

When viewing TV and movie clips, for example, neuroimaging evidence suggests that adult brains differentially track low- and higher-level features (Chang et al., 2022 [↗](#)). Temporally coarse-grained predictions originating in the default mode network inform temporally fine-grained predictions in primary auditory and motor areas (Baldassano et al., 2017 [↗](#); Chang et al., 2022 [↗](#); Hasson et al., 2015 [↗](#)). In infants, these processes are thought to be more rudimentary. Infant brains represent only longer events, even in early visual regions, with no time-scale hierarchy (Yates et al., 2022 [↗](#)). Whereas younger infants' gaze is better predicted by layers of a neural network model corresponding to lower-level areas of the ventral visual stream, older infants' gaze was better predicted by higher-level layers (Kiat et al., 2022 [↗](#)). Over time, information increasingly becomes integrated over more coarse-grained spatial and temporal scales (Pempek et al., 2010 [↗](#)). As multiple physical aspects of an event reach an infant, they must be related to each other for the infant to model a complete representation of that event (Bahrick & Lickliter, 2000 [↗](#); Dionne-Dostie et al., 2015 [↗](#)).

1.1 The Role of Caregivers in Scaffolding Infant Attention

However, these previous studies do not account for the fact that most early real-world experiences are embedded in social interaction (Carretero & Español, 2016 [↗](#); Español & Shifres, 2015 [↗](#)). Traditional approaches use experimental tasks that conceptualise attention as a passive response to incoming sensory information (L. Smith & Gasser, 2005 [↗](#); Wass & Jones, 2023 [↗](#); Wass, 2014 [↗](#); Wass & Goupil, 2022 [↗](#)). But in reality, infants actively modify their environments through their interactions, shaping the information they receive based on their own behaviour and experiences (Anderson et al., 2022 [↗](#); Mendez et al., 2024 [↗](#); L. Smith & Gasser, 2005 [↗](#); Wass & Goupil, 2022 [↗](#)). This is true of children's interactions with physical objects and features in their environment (Anderson et al., 2022 [↗](#), 2024 [↗](#)) but it is especially true of their early social interactions.

Extensive evidence already shows that caregivers shape infants' attention and learning by adapting their behaviour to align with the developmental needs of the infant (Brennan et al., 2010 [↗](#); Schick et al., 2022 [↗](#); Tippenhauer et al., 2020 [↗](#)). One well-studied example is the use of infant-directed (ID) speech, which is characterized by exaggerated acoustic features such as higher pitch, greater pitch variability, and slower amplitude modulation (M. Cooke et al., 2014 [↗](#); Hilton et al., 2022 [↗](#)). These features, although not completely universal, act to capture and maintain infant attention effectively while facilitating language learning (Nencheva & Lew-Williams, 2022 [↗](#); Räsänen et al., 2018 [↗](#)). Additionally, caregivers simplify semantic complexity in their speech,

adjusting it to match the infant's vocabulary development (Schwab et al., 2018; Schwab & Lew-Williams, 2016) and further supporting word learning. Caregivers also modify their facial expressions and hand movements to maintain infant engagement and visual attention (Chong et al., 2003; Kliesch et al., 2022; Stern, 1974; van Schaik et al., 2020). These adaptations are thought to be constrained by diverging need (Chater & Vitányi, 2002; Woźniak & Knoblich, 2022): predictable elements help infants integrate upcoming stimuli, while novel stimuli capture attention by introducing uncertainty (Kidd et al., 2012, 2014; Meyer et al., 2023; Labendzki et al., pre-print).

1.2 “Alive” joint attention

Most current quantitative research on infant attention has focused on global differences, such as average comparisons between infant-directed versus adult-directed communication. These approaches miss the dynamic, reciprocal, “alive” (Fogel & Garvey, 2007) nature of social interactions. Although countless studies have used observer ratings to measure maternal sensitivity and dyadic mutuality (J. E. Cooke et al., 2022; Feldman, 2007; Murray et al., 2016), relatively few studies have quantitatively observed how infants and caregivers continuously respond and adapt to each other's behaviours in a bidirectional exchange (Beebe et al., 2016; Feldman, 2007; Jaffe et al., 2001).

Previous research has shown that mothers dynamically adjust both the pitch of their speech and their modulation patterns (rate of change of pitch) based on infants' attentional states (Phillips et al., 2023; Reisner et al., 2024; N. A. Smith & Trainor, 2008). We also know that caregivers modulate their gaze contingent on infant behaviours (Perapoch Amadó et al., 2025), and that mothers' object labelling and handling are often contingent on infants' gaze (Goupil et al., 2024; Sun & Yoshida, 2022). From previous research we also know that object naming enhances infant attention during joint play (Mendez et al., 2023; Sun & Yoshida, 2024). This creates a shared history of interaction, enabling both partners to predict and adapt to each other's actions, ultimately fostering a dynamic communication system (Bruner, 1974; Fogel et al., 1992; Gratier & Magnier, 2012; Hasson & Frith, 2016; Malloch & Trevarthen, 2009; Murray, 2014; Ravreby et al., 2022; Vygotsky et al., 1978).

As yet, though, most previous that has examined dynamical processes during real-time caregiver-infant interactions has concentrated on examining different individual features in isolation (Beebe et al., 2016; Ham & Tronick, 2009; Lavelli & Fogel, 2013; Phillips et al., 2023). No previous research has examined how these reciprocal interactive influences operate across multiple hierarchical layers, ranging from low-level salience cues (such as physical movement, and vocal pitch and amplitude fluctuations) through to higher-level features that represent meaning and context. Understanding this is crucial from both a practical perspective, for improving interventions that target interaction dynamics to improve long-term infant outcomes (e.g. Murray et al., 2016); and from a theoretical perspective, for enriching our understanding of how multimodal and hierarchical features interactively influence social attention, and how these influences change and develop with age.

1.3 Current Study

The present study examines how infant-caregiver interaction change between 5 and 15 months, which is the period when the capacity for infant-led joint attention is thought to emerge (Mundy et al., 2009). We manually coded the gaze of mothers and their infants as they played together with toys (see 2.2 for more details) and calculated a range of lower-level and higher-level features of the interaction. Lower-level features included ‘spectral flux’, i.e. the instantaneous change in audio spectral content, which was calculated separately for ambient noise and for sections where the mother was vocalising; and differentials (i.e. rate of change) of maternal face and hand movement, which drive low-level visual and auditory salience (Itti & Baldi, 2009). Higher-level features operate over longer timescales and involve the integration of low-level features into information

that is cognitively meaningful. These included information rate (i.e., the rate of meaningful data transmitted per time unit), semantic surprisal (i.e., log probability of the upcoming word given the preceding words), toy naming, and facial novelty (i.e., the predictability of mothers' facial expression given previous expression) (see 2.2 [for equations and formal definitions](#)). The distinction presented here between lower- and higher-level features is a practical simplification; in reality, lower- and higher-level features exist along a continuum (Gwilliams et al., 2024 [for equations and formal definitions](#); Heilbron et al., 2022 [for equations and formal definitions](#)).

We had three main predictions. **Prediction 1: Maternal behaviours and infant attention.** Across the age groups, mothers' behaviour will be tightly coupled to changes in infant attention to objects over modality (i.e., audio and visual) and levels (low and high order). **Prediction 2: Age-dependent effectiveness of features.** Lower-level features will be more effective in both capturing and maintaining younger infant's attention to objects, while higher-level features will be more effective with older infants. **Prediction 3: Changing temporal dynamics.** The temporal relationship between fluctuations in maternal features and infants' attention to objects will change with age. At 5 months, lower-level variables will more frequently precede infant attention. However, as infants increasingly take the lead in play, we predict that changes in higher-level features of the mother's behaviour will more often follow the infant's attention to objects.

To explore these three predictions, we first calculated the mean and median for all the features of the interaction we studied. Following this, we built a cross-correlation matrix to understand how our behavioural variables (e.g. the lower-level and higher-level features of the interaction) inter-relate. Finally, we performed cross-correlations to examine the temporal relationship between each lower- and higher-order feature and the infant's attention to objects.

For our primary analyses, infants' attention to objects was conceptualized as the 'on-task' behaviour reflecting active engagement with the central goal. In addition, however, because our tabletop play setting allowed for three gaze locations (object, partner, inattentive) we also wished to examine the possibility that decreases in infant's attention to objects might be more readily explained as increases in infants' attention to the mother's face. To test this, we also include analyses in the Supplementary Materials in which we performed the same cross-correlations but with infants' looks to the mother's face rather than to objects as the dependent variable.

2. Materials and methods

2.1 Participants

Participants were typically developing infants and their mothers. Only mothers were included because of practical difficulties in recruiting sufficient fathers to provide a gender-matched sample. The catchment area for this study was East London, including boroughs such as Tower Hamlets, Hackney and Newham. Further demographic details on the sample are given in **Table S1** [for equations and formal definitions](#).

Participants were recruited postnatally through advertisements at local baby groups, local preschools/nurseries, community centres and targeted social media campaigns aimed at all parents in the area, from databases of prior projects and via word-of-mouth. Informed consent and authorisation for publication has been obtained by the caregivers featured in **Fig 1** [for equations and formal definitions](#) and in **Fig.S1** [for equations and formal definitions](#). Ethical approval was obtained from the University of East London ethics committee (application ID: ETH2021-0076).

Initial exclusion criteria include complex medical conditions, known developmental delays, prematurity, uncorrected vision difficulties and parents below 18 years of age. Further exclusion criteria as well as final numbers of data included in each of the analyses for both samples are



Figure 1

Illustration of analyses with the central figure showing all low- and high-level features synced.

A) Spectral flux being computed as the differences between consecutive spectrum (green area between the current and previous spectrogram). B) Hand movement is computed as the average distance travelled by the right and left hands. C) Face movement is computed as the sum of derivative of the eye-to-brow distance (blue dot on the eye to red dot on the eyebrow) and the mouth opening distance (blue dot on the upper lip to red dot on the lower lip). D) Object naming was obtained using the automatic transcription and a query for the specific toys present during the interaction. E) Semantic surprisal. For each word, a probability distribution was obtained using GPT2 prompted with the previous words, the semantic surprisal is the log probability of the observed word. F) Information rate. For each word, a cumulative complexity (upper) is computed using lossless compression algorithms, taking the derivative gives the information rate (lower). G) Facial novelty. For every frame, the facial expression is estimated, and an information distance is computed using the Kullback-Leibler between consecutive frames.

summarised in **Table S2** [↗](#). The final samples included 33 5-month-old infants (15 females) and 34 15-month-old infants (17 females) and their mothers. Data were analysed in a cross-sectional manner. Average age for infants was 5.23 months (std= 0.5) and 15.75 months (std= 1.15) respectively. Average age for mothers was 35.31 years (std = 3.9, N= 29) at 5 months and 37.41 years (std= 3.82, N= 28) at 15 months. Two previous studies on joint attention and imitation have already reported analysis of the same participant cohort (Perapoch Amadó et al., 2025 [↗](#); Viswanathan et al., pre-print [↗](#)) although it is the first time that the rest of the data (i.e. lower- and higher-order variables) are analysed and reported.

2.2 Experimental design

Mothers and infants were seated facing each other on opposite sides of a table. Infants were seated either in a highchair or on a researcher's lap, within easy reach of the toys (see **Fig S1A** [↗](#)). At the beginning of the joint play session, a researcher placed the toys on the table and asked the mothers to “play with their infants just as they would at home”. During the play session, researchers stayed behind a divider out of view of both the mother and the infant. The same three toys were used for each age group (see **Fig S1D** [↗](#)). The average duration of the joint play interactions was 5.06 minutes (std= 1.37) at 5 months and 6.17 minutes (std= 1.64) at 15 months (**Fig S1C** [↗](#)). Average duration differed significantly between 5- and 15-months ($t(65) = -3.014$, $p=0.003$). Given that the analyses conducted here are relative to the duration of each interaction or on specific events (e.g. infant looks at objects), variations in interaction durations should not be an issue.

The interactions were filmed using three Canon LEGRIA HF R806 camcorders recording at 50 frames-per-second (fps). Two cameras were placed in front of the infant, one on each side of the mother, and another one was placed in front of the mother, just behind the right side of the infant. All cameras were positioned so that the infant's and the mother's gaze, as well as the three toys placed on the table, were always visible (see **Fig S1A** [↗](#)). Microphone data were also collected using two wireless omni-directional Lavalier microphones recording at 44.1kHz in wav format, one attached to the mothers' clothing and the other to the infant's highchair.

Of note, brain activity was also recorded from both the infants and their mothers, at both ages, using a 64-channel BioSemi gel-based ActiveTwo EEG system. However, this data is not included in the current manuscript.

2.3 Data processing

2.3.1 Synchronisation of the different datasets

The cameras pointing at the participants were synchronised via radio frequency (RF) receiver LED boxes attached to each camera. The RF boxes received trigger signals from a single source (computer running Matlab) at the beginning and end of the play session and concurrently triggered light pulses to LED lights visible to each camera, along with an audible beep. The synchronisation of the video coding was conducted offline by aligning the times of the LED lights of the three cameras and checking that the durations matched. The audio data from the two microphones was synchronised from the start using the Zoom H4N PRO Handy Recorder, which enables simultaneous recording. Finally, Adobe Premiere Pro was used to synchronise the video with the audio data, allowing us to align the two datasets in time and estimate the lag between them.

2.3.2 Gaze behaviour coding and processing

The looking behaviour of the infants and their mothers was manually coded offline on a frame-by-frame basis, at 50fps. The start of a look was the first frame in which the gaze was static after moving to a new location. The following categories of gaze were coded: looks to objects (focusing on one of the three objects), looks to partner (looking at their partner), inattentive (not looking to any of the objects nor the partner) and uncodable (see [Fig S1B](#)). Uncodable moments included periods where: 1) their gaze was blocked or obscured by an object and/or their own hands, 2) their eyes were outside the camera frame, and/or 3) a researcher was within the camera frame.

To assess inter-rater reliability, ~22% of the data (15 datasets) were double coded by a second coder and both Cohen's kappa and observed agreement were calculated. There was substantial agreement ($\kappa = 0.628$, $\text{std} = 0.134$; Kappa error = 0.005, $\text{std} = 0.001$; observed agreement = 0.751, $\text{std} = 0.092$) ([Landis & Koch, 1977](#)). Looking behaviour data was then processed such that any look preceding and following an “uncodable” period was excluded from further analyses. Similarly, both the first and the last look of every interaction were also excluded from further analyses.

2.3.3 Correlation analyses and calculation of significance

To calculate the associations between different features of the interaction we first performed a standard correlation analysis between all variables ([Fig 3](#)). Next, we conducted cross-correlation analyses to examine the dynamic associations between interaction features and infant attention (3.3) (i.e., whether one variable leads and the other follows) ([SM Fig 5](#)). First, we first converted the infants' looking behaviour data into binary arrays of ones (looking at an object) and zeros (not looking at an object) ([Wass et al, 2019](#); [Phillips et al., 2023](#)). Following this, we linearly detrended all the time series datasets and calculated the cross-correlations between all the different features of the interaction (3.2), and between these and infant attention (3.3) separately.

Because cross-correlations are not time locked to specific moments (i.e. onset of an infant look to object) but instead are conducted on two time series (e.g. infant looks to objects and information rate) as a whole (see [Fig 1](#)), the strength of the overall correlation is weakened by the fact that periods of expected stronger correlation are balanced by weaker correlations where we would not expect any correlation at all ([Xu et al., 2020](#)). This can lead to very small correlation coefficients (e.g. around $r = \pm 0.05$), albeit sometimes significant when compared against the permuted correlations (see below).

To assess whether the results from the cross-correlations were significantly different than chance we generated permuted data and compared it against the observed data using Cluster-Based Permutation (CBP) tests. To generate permuted data, we computed the cross-correlation between 500 random combinations of time series data from different dyads. For example, for each feature, the time series data of one participant (e.g. information rate from dyad 1) was randomly paired with the time series data of another participant (e.g. infant looking behaviour from dyad 13). We repeated this 500 times and then performed a CBP test to examine significant differences between the results from the observed (i.e. real) and the permuted data.

The CBP test statistic was calculated using a function from FieldTrip ([Maris & Oostenveld, 2007](#)) called “ft_timelockstatistics”. This nonparametric framework allowed us to both control for the multiple comparison problem that arises from the fact that the effect of interest is evaluated many times (e.g. changes in mothers' spectral flux around infant looks to objects), and to reduce the potential for false negative effects ([Meyer et al., 2021](#)).

Please refer to [Fig S4](#) for a visual guide on interpreting findings from a cross-correlation analysis, illustrating the interpretation of positive and negative cross-correlation values across forward and backward time-lags. Of note, we refer to ‘forward lags’ as positive lags and ‘backward

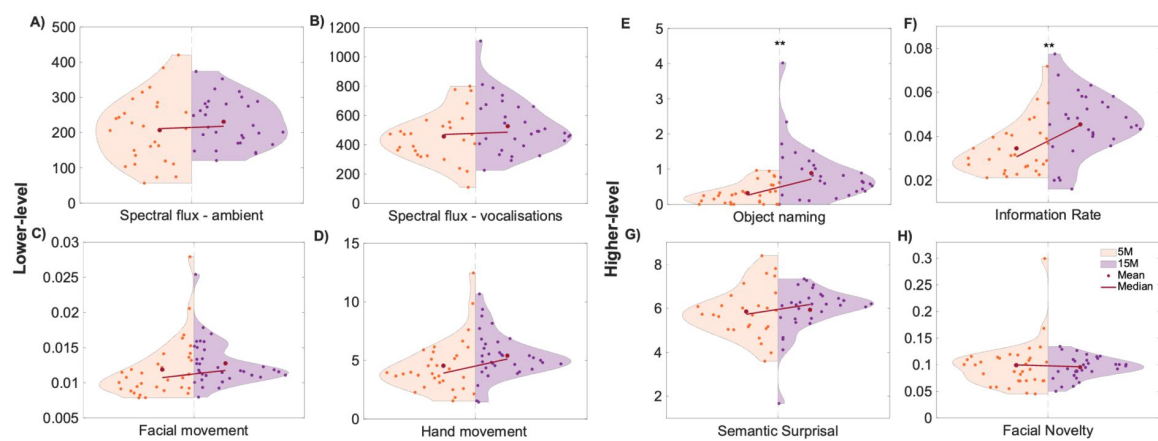


Figure 2.

Descriptive analyses on lower- and higher-level features.

Violin plots showing the average at a group level of lower-level: Spectral flux - ambient (A); spectral flux - vocalisations (B); facial movement (C); hand movement (D), and higher-level features: object naming (E); information rate (F); semantic surprisal (G); facial novelty (H). Individual dots represent the data for each participant, in orange is data at 5 months, and in purple is data at 15 months. Red dots are showing the mean and red lines are showing the median. Asterisks indicate significance (* = $p\text{-adj} < 0.05$, ** = $p\text{-adj} < 0.01$, *** = $p\text{-adj} < 0.001$).

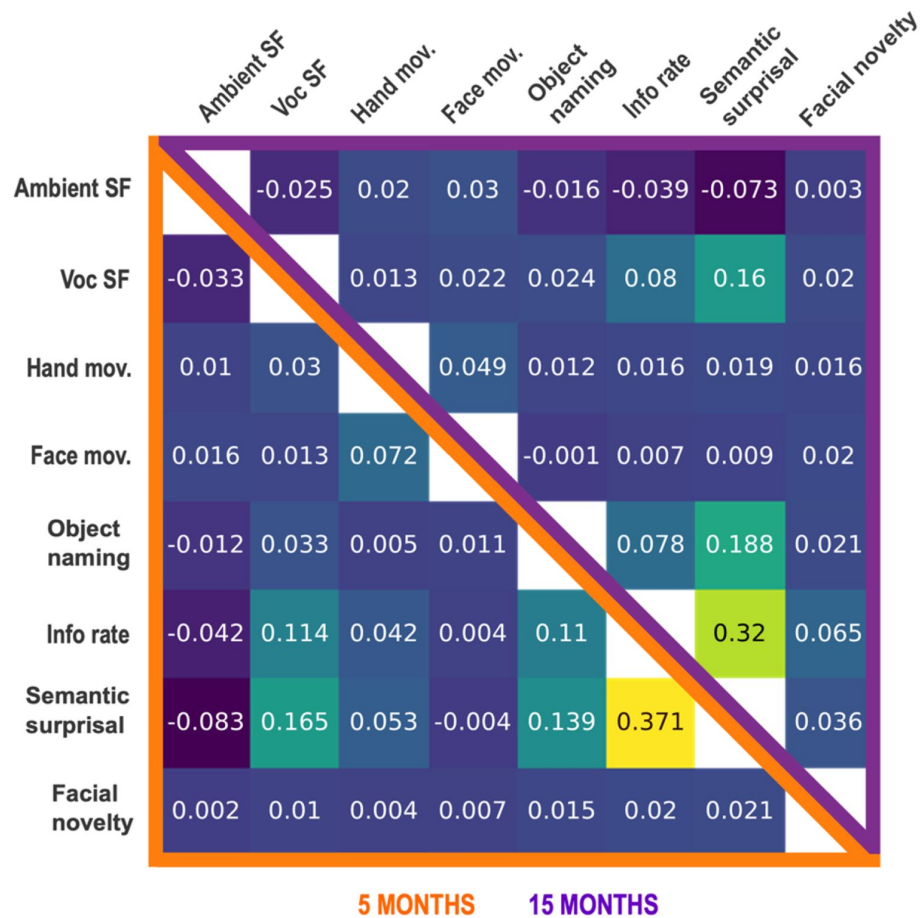


Figure 3.

Correlation matrix between all features of the interaction at 5 and 15 months.

Each square shows the correlation between two maternal behavioural variables with a zero lag. The bottom-left triangle shows correlation values for the 5 months visit, and the top-right triangle shows correlation values for the 15 months visit.

lags’ as negative lags to avoid redundancy with the terms ‘positive’ and ‘negative’ used to describe correlation values.

2.3.4 Calculation of features of the interaction

In this section, we explain how both lower and higher-level features were calculated. Of note, these variables were all resampled to match the video sampling rate of 50Hz.

2.3.4.1. Calculation of lower-level features

2.3.4.1.1 Spectral Flux

Spectral flux (SF) is a measure of acoustic change over time. In this study, we opted to separate spectral flux of the ambient background (i.e., other environmental noises such as claps, toys banging on the table, etc) from spectral flux of the mothers’ vocalisations to better analyse the distinct acoustic properties and influences of each on infant attention (see **Fig 1A and 1B**). We computed spectral flux as the sum of absolute differences over frequency of successive short-time Fourier transform (Müller, 2015).

$$SpectralFlux(t) = \sum_{0\text{ Hz}}^{3000\text{ Hz}} |X(t-0.02; f) - X(t; f)|$$

where $X(t;f)$ is the spectral amplitude at time t and frequency f computed over a 0.05sec window with 0.02sec hop-size.

Spectral flux was chosen as a metric for low-level auditory saliency, as it measures both amplitude and frequency changes over time and reflects acoustic differences between consecutive time points, both important salient auditory features (Huang & Elhilali, 2017; Kothinti et al., 2021). These features are modulated in Infant Directed Speech (IDS) (Cooke et al., 2014; Leong et al., 2017) and signal short term novelty in the audio stream (Müller & Chiu, 2024). Furthermore, spectral flux has been found to be a better predictor of neural entrainment to music, as it also measures changes in the frequency domain, compared to amplitude envelope only (Weineck et al., 2022). This approach allows us to have a holistic approach by measuring all possible spectral changes in naturalistic speech.

We then split spectral flux into two complementary streams using OpenAI’s Whisper (Radford et al., 2023), an open-source automatic speech recognition system. The general architecture of the classifier as well as the process followed to train the model are presented in more detail in Radford et al., (2023). WhisperX (Bain et al., 2023) was also used to obtain word-level onsets and offsets with millisecond precision. From there, we split the spectral flux for both vocalisations and ambient environmental sounds (i.e., all periods that occurred outside of the periods where WhisperX detected speech) separately.

2.3.4.1.2 Face movement

Face movement measures low-level facial movements, particularly the coordination of eyebrow and mouth movements. We used the “whole body” model from MMPose toolbox (mmpose author 2020) to extract two dimensional coordinates of mothers’ body features in every frame. The raw 2D coordinates were cleaned using a series of low-pass filters designed to reject movements that were impossibly fast. We then interpolated those short missing segment and low-pass filtered the time series to reduce the jitteriness from the feature estimation. To calculate the “Face movement” variable, we first averaged the distances between the left and right eye-to-brow measurements and then averaged this result with the mouth opening measurement. This approach captures low-

level facial movements, which are considered salient during mother-infant interactions (Biringen, 1987 [↗](#)), in particular mouth movements (Lewkowicz & Hansen-Tift, 2012 [↗](#); Zhang et al., 2021 [↗](#)) and eyebrow movements (de Klerk et al., 2018 [↗](#); Isomura & Nakano, 2016 [↗](#)).

Distances between the eye-to-brow as well as mouth opening were calculated using the Euclidian distance between the 2D coordinates of the centre of the eye, centre of the brow, and upper and lower lips (see **Fig 1C** [↗](#)). These distances were then normalised by the nose length at each frame to account for dynamic distance between the mothers' head and the camera.

2.3.4.1.3 Hand movement

Our hand movement measure reflects the average distance travelled by the left and right hands. We use the same pipeline as for the face movement analysis ("whole body" model from MMPose with in-house cleaning) to extract the distance travelled over time by the left and right hands and then computed their average. We took the distance travelled by the hands as an index of low-level movement activity within the computed time window. The distance travelled was computed over 50 consecutive samples (1 second) and is proportional to speed.

2.3.4.2 Calculation of higher-level features

2.3.4.2.1 Object naming

Here we created a binary array where ones represented the name of the toys (i.e. panda, book and rattle; **Fig S1D** [↗](#)) and zeroes indicate the absence of these words. To obtain the onsets and offsets of each toy object we used Open AI's Whisper (Radford et al., 2023 [↗](#)) and WhisperX (Bain et al., 2023 [↗](#)) (explained above in 2.3.4.1.1).

2.3.4.2.2 Information Rate

Information rate is a measure of the amount of new (uncompressible) information per unit of time. It was computed as the derivative of the cumulative compression size (Schmidhuber, 2009 [↗](#)). In our case it was computed at a word level and can be expressed as:

$$IR(w_i) = \text{compression}(w_i; w_{i-1}; \dots; w_1) - \text{compression}(w_{i-1}; \dots; w_1)$$

With w_i being the i^{th} word and $\text{compression}(w_i)$ being the size of the losslessly compressed representation of that word. The lossless compression technique can track redundancy in text sequences in hierarchically increasing n-sequences. Compressing text in a cumulative window with PyLZMA, a Lempel-Ziv-Markov algorithm with dynamical dictionaries (Bauch et al., 2015 [↗](#)), resembles how adults make predictions about upcoming events (Hasson et al., 2015 [↗](#); Schmidhuber, 2009 [↗](#)). The upcoming words are predicted from the integrative posterior, i.e. the words (or sequences of words) that have already come. For instance, as more words are introduced the size of the description might increase. New words (i.e. words that have not already come before) would increase the description's size by a larger degree when compared to the introduction of words that have already come. By taking the derivative of that increasing size, we effectively compute the rate at which new information is introduced. This integrative compression process can be compared to predictive processing (Schmidhuber, 2009 [↗](#)) and to cognitive constructivism (Chater & Vitányi, 2002 [↗](#); Wolff, 2014 [↗](#), 2019 [↗](#)) where a pattern-seeking agent compresses incoming data into a modelled representation using redundancies from the data.

2.3.4.2.3 Semantic surprisal

Semantic surprisal measures the linguistic probabilistic inference: it quantifies how surprising each upcoming word is given their context. This approach was chosen as it shows correlation with reading time and the N400 (S. L. Frank et al., 2015 [↗](#); Shain et al., 2024 [↗](#)) and was computed for

each word as the negative log probability of that word given all previous words (Willems et al., 2016). Specifically, this was conducted using the GPT-2 large language model, a generative transformer model that predicts the most probable next words given a text, along with their respective probabilities. We then retrieved the probability of the observed upcoming word and computed the negative log₂ of that probability as the semantic surprisal (Shannon, 1948). The word was then added to the context window, and the prediction process was repeated for the next upcoming word (see Fig 1.E). Importantly, we prompted every interaction with the same sentence presenting the experiment to prevent the model from being surprised when it encountered normally unusual or unexpected words such as ‘panda’ (“The following text is about a mother and her infant playing together, as they would at home, with a toy panda, a book and a rattle, while wearing EEG hats and facial electrodes”).

$$\text{Semantic Surprisal}(w_i) = -\log_2(P(w_i | w_{i-1}; \dots; w_1))$$

2.3.4.2.4 Facial novelty

Our measure of facial novelty reflects an estimate of changes in the facial expressions of the mother over time. To calculate these, we used an open-source Python library, the ‘Facial Expression Recognition using Residual Masking Network’ (Pham et al., 2020). This classifier gives probabilities for seven canonical facial expressions (happy, sad, surprise, angry, disgust, fear, neutral) using residual neural network, a machine learning model that shows high generalisability due to “skip connections” that allows the model to ignore irrelevant features and focus on informative features for the task. On adult faces the model performs with 76.82% accuracy compared to manually labelled expressions (Pham et al., 2020). To measure the changes over time we computed the Kullback-Leibler divergence between consecutive frames.

$$\text{FacialNovelty}(t) = \sum_{x \in \text{exp}} E_{x,t} \log_2 \left(\frac{E_{x,t}}{E_{x,t-1}} \right)$$

Where $E_{x,t}$ is the expression probability for the expression x distribution at time t . The Kullback-Leibler divergence can be interpreted as the information gained from updating $E_{x,t-1}$ to $E_{x,t}$, e.g. if the divergence is high, it means that the most recent facial distribution vector required a lot of information to be updated, i.e. the facial expression was novel compared to the previous frame (see Fig 1). We classified this as a higher-level feature as it the frame-wise difference between the last, and most abstract layer of neural network (Giordano et al., 2023; Kiat et al., 2022; Kothinti & Elhilali, 2023; Lecun et al., 2015; Zeiler & Fergus, 2013), and because we chose an information metric measure that has been used to model infant attention (M. C. Frank et al., 2009; Poli et al., 2020) to measure the model updating between consecutive frames.

3. Results

3.1 General descriptives

First, we explored between- and within-age group differences in how many times per minute infants and mothers engaged in looks to objects (see SM, Fig S2A) and for how long these episodes lasted on average (see SM, Fig S2B). We found no differences between age groups in the average look duration to objects or counts for either infants or mothers. However, mothers looked more frequently to objects ($t_{5M}(61) = -3.01$, $p_{5M} = 0.004$; $t_{15M}(63) = -6.34$, $p_{15M} < 0.001$) but for shorter durations ($t_{5M}(61) = 5.43$, $p_{5M} < 0.001$; $t_{15M}(63) = 8.47$, $p_{15M} < 0.001$) compared to infants at both ages. The same analyses on looks to partner are presented in the SM (Fig S2C and D).

Second, we quantified the amount of caregiver speech during the joint play interactions at 5 and 15 months (see [SM Fig S3](#)). The amount of speech was greater at 15 months compared to 5 months ($t(52) = -2.43, p = 0.018$).

Finally, we calculated the overall mean and median values for all the lower- and higher-level features. This allowed us not only to see how these variables are distributed across participants but also to conduct age group comparisons. To assess significant differences across age groups, we first tested for normality using MATLAB's "vartestn" function. For normally distributed data, we applied two-sided t-tests, and for non-normally distributed variables, we used two-sided Wilcoxon rank-sum tests. We adjusted for the family wise false positive discovery rate using the false discovery rate (FDR) method, using the Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) with an alpha set at 0.05. We report adjusted p values (p-adj) that have been corrected for multiple comparisons in [Table S3](#) after each uncorrected p value. We observed that both the averages for "Information Rate" and "Object naming" increased significantly between 5 months and 15 months (see [Table S3](#)). The rest of the variables did not change significantly from 5 to 15 months.

3.2 Correlations between all features of the interaction

We performed a standard correlation analysis across all eight lower- and higher-level features to examine the associations between all the different interaction features. [Figure 3](#) shows the correlation matrix at 5 and 15 months (see [Fig. 3](#)). We also conducted a cross-correlation analyses to capture more dynamic associations (see [SM Fig 5](#)).

Overall, both analyses revealed that most variables were interrelated, as evidenced by significant correlations observed across multiple feature pairs ([SM Fig 5](#)). While the strength and timing of these correlations vary, most of them exhibit significant positive associations, particularly around lag 0, suggesting a high degree of temporal coordination (see [Fig 3](#) for correlation values at lag 0, and [SM Fig 5](#) for cross-correlation results). For example, when 'Spectral Flux of vocalisations' is high, 'semantic surprisal', 'Information rate' and 'object naming' are also high.

Conversely, the only pairs exhibiting negative correlations were the ones involving 'spectral flux of ambient noise' and the speech related variables ('spectral flux of vocalisations', 'object naming', 'information rate' and 'semantic surprisal'). This relationship is likely due to the way these variables are calculated, as they are inherently interdependent. The spectral flux of ambient noise and the spectral flux of vocalisations or information rate are measured in mutually exclusive contexts: when mothers are vocalising, spectral flux of ambient is set to zero and when spectral flux of ambient is being measured, it indicates that mothers were not vocalising. As a result, the observed pattern is a byproduct of our methodology rather than an intrinsic relationship between the variables.

The variables 'facial movement' and 'facial novelty', although both measuring the dynamics of facial feature, showed correlation values below .02, validating the theoretical differences described in the method section. Interestingly, there were few differences observed between 5 and 15 months.

3.3 Cross-correlations between features of the interaction and infant attention to objects

For our primary analyses we conducted cross-correlations to examine whether certain features of the interaction, most of them related to maternal behaviours, forward-predicted changes in infant attention to objects, or vice versa. Our predictions were that mothers would respond to decreases in infant attention by increasing their own lower-level salience (lower-level features forward-

predicts infant attention); but then, when the infants' attention is re-engaged, they would downregulate salience and upregulate higher-order semantic features (infant attention forward-predicts higher-level features).

Our findings are presented below, organized by each variable. The p-values resulting from statistical tests were adjusted separately for the lower-level and higher-level groups using the Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995 [↗](#)) with an alpha set at 0.05. The adjusted p-values are reported as 'p-adj' in the main text alongside the corresponding uncorrected p-values. Unless otherwise specified, all reported findings reflect bidirectional relationships (i.e., if an increase in X leads to a decrease in Y, the opposite - where a decrease in X leads to an increase in Y - also holds). Please refer to **Fig S4** [↗](#) for a visual guide on interpreting findings from a cross-correlation analysis.

Since one important question in interpreting our findings is whether, instead of paying attention to objects, infants are instead paying increased attention to their parents, we have also presented in the SM (**Figs S6** [↗](#) and **S7** [↗](#)) the exact same set of analyses but examining instead the associations between features of the interaction and infant attention to mother.

3.3.1 Lower-order features

3.3.1.1 Spectral flux - ambient

As expected, we observed significant negative correlation values at $t=0$, suggesting that less ambient spectral flux associated with more infant attention to objects. When we introduced a time lag between the variables, we found that this negative association was significant from $t=-1.1/-0.86\text{sec}$ (infant precedes ambient SF) to $t=3.06/3.54\text{sec}$ (ambient SF precedes infant) at 5 months ($p=0.008$, $p\text{-adj}=0.013$) and at 15 months ($p=0.009$, $p\text{-adj}=0.013$) respectively (**Fig 3A, 3B** [↗](#)). The significance of both findings is stronger for positive time-lags indicating that, overall, reductions in the ambient spectral flux tended to forwards-predict increases in infant's subsequent attention to objects (or that increases in ambient spectral flux forward predicted decreases in infant's object attention) at both ages (**Fig 3A, 3B** [↗](#)).

3.3.1.2 Spectral flux - vocalisations

We found negative correlation values at $t=0$, suggesting that maternal speech containing less spectral flux associates with more infant attention to objects. At 5 months, this relationship was significant during two temporal windows: from $t=-2.1\text{sec}$ (infant precedes mother) to $t=1.24\text{sec}$ (mother precedes infant) ($p=0.015$, $p\text{-adj}=0.017$), and from $t=6.54\text{sec}$ to $t=9.52\text{sec}$ ($p=0.022$, $p\text{-adj}=0.022$) (**Fig 3C** [↗](#)). This suggests that, when 5-month-old infants focus on objects, mothers reduce the acoustic variability of their vocalisations, or conversely, when infants' attention to objects decreases, mothers increase the acoustic variability of their vocalisations (backward time-lag) (see **Figure S4** [↗](#)). Additionally, we also found that an increase in mothers' acoustic variability was followed by a reduction in infants' attention to objects, or *vice versa*, when mothers decrease their acoustic variability, infants' attention to objects increased (forward time-lag). At 15 months, instead, these negative associations were present at forward time-lags but not statistically significant (**Fig 3D** [↗](#)).

We considered the possibility that an increase in the spectral flux of maternal vocalisations might be associated not with an increased likelihood of looking at the object, but rather with an increased likelihood of looking at the mother. However, we found no support for this SM (**Fig S6C and D** [↗](#)).

3.3.1.3 Facial movement

We found negative correlations at time $t=0$, suggesting that increased maternal facial movement associates with less infant attention to objects, at both 5 and 15 months. This was significant from $t=-3.92/-3.46\text{sec}$ to $t=2.48/3.06\text{sec}$ at 5 ($p=0.006$, $p\text{-adj}=0.013$) and 15 months ($p<0.001$, $p\text{-adj}=0.007$) respectively (**Fig 3E**, [F](#)). This suggests that when infants focus on the objects, mothers make fewer facial movements, potentially removing stimuli from an infant's peripheral vision. Conversely, when mothers increase facial movements, infants tend to look less at the objects (**Fig 3E**, [F](#)). As expected, when we looked at the relationship between maternal facial movement and infants' looks to their mother, we found the opposite pattern (see **SM**, [Fig S6E](#), [F](#)). When infants look at their mother, there is an increase in maternal facial movement; and when mothers increase facial movements, infants look at their mothers more. Again, this pattern was observed at both ages. This relationship is independent of the relationship with vocal spectral flux documented in 3.3.1.2 (see **Fig 3**, [Fig S5](#)) because the direct association between the two variables is negative, but the two variables are each negatively associated with infant attention.

3.3.1.4 Hand movement

We found negative correlations at time $t=0$, suggesting that increased maternal hand movement associates with less infant attention to objects, at both 5 and 15 months. This was significant from $t=-3.6/-8.02\text{sec}$ to $t=1.52/1.02\text{sec}$ at 5 ($p=0.01$, $p\text{-adj}=0.013$) and 15 months ($p=0.003$, $p\text{-adj}=0.012$) respectively (**Fig 3G**, [H](#)). The significance of both findings is stronger at backward time-lags indicating that, overall, when an infant focuses their attention on an object, it leads to a decrease in the mother's hand movements (or that a decrease in the infants' attention to an object leads to an increase in the mother's hand movements) (see **Fig 3G**, [H](#)). Facial movement and hand movement were themselves positively correlated (see **Fig 3**, [Fig S5](#)), so the similar associations with attention likely pick up common variance across multiple variables.

3.3.2 Higher-level features

3.3.2.1 Object naming

No association between object naming and infant attention was observed at 5 months (**Fig 4A**). At 15 months, instead, we observed a positive significant correlation from $t=0.24$ to $t=2.6\text{sec}$ ($p=0.032$, $p\text{-adj}=0.032$), indicating that, when mothers name objects, it associates with subsequent increases in infants' attention towards the named object (**Fig 4B**). These associations may be due to the shared variance between object naming and low-level features documented previously (see **Fig 3**, [Fig S5](#)). However, the fact that the associations with low-level features were consistent at both ages, whereas the association between object naming and attention was present only at 15 months, partially precludes this possibility.

3.3.2.2 Information rate

We observed a significant positive correlation between information rate and infant attention where increased maternal information rate associated with increased infant attention to objects. This was significant from $t=2.46$ to 5.56sec at 5 months ($p=0.023$, $p\text{-adj}=0.032$; **Fig 4C**), but non-significant at 15 months (**Fig 4D**). These findings are stronger at forward time-lags indicating that, overall, increases in maternal information rate more often precede infant attention to objects than follow it. Interestingly and conversely, we observed the opposite pattern for infant attention to the partner, where maternal information rate tended to precede a decrease in infant attention to partner. This pattern was present at both ages, though it was only statistically significant at 5 months (see **SM**, [Fig S7C](#), [D](#)). This association is independent of the relationship documented between information rate and vocal spectral flux (see **Fig 3**, [Fig S5](#)) because those two variables are positively associated whereas the latter is positively associated with infant attention and the latter negatively.

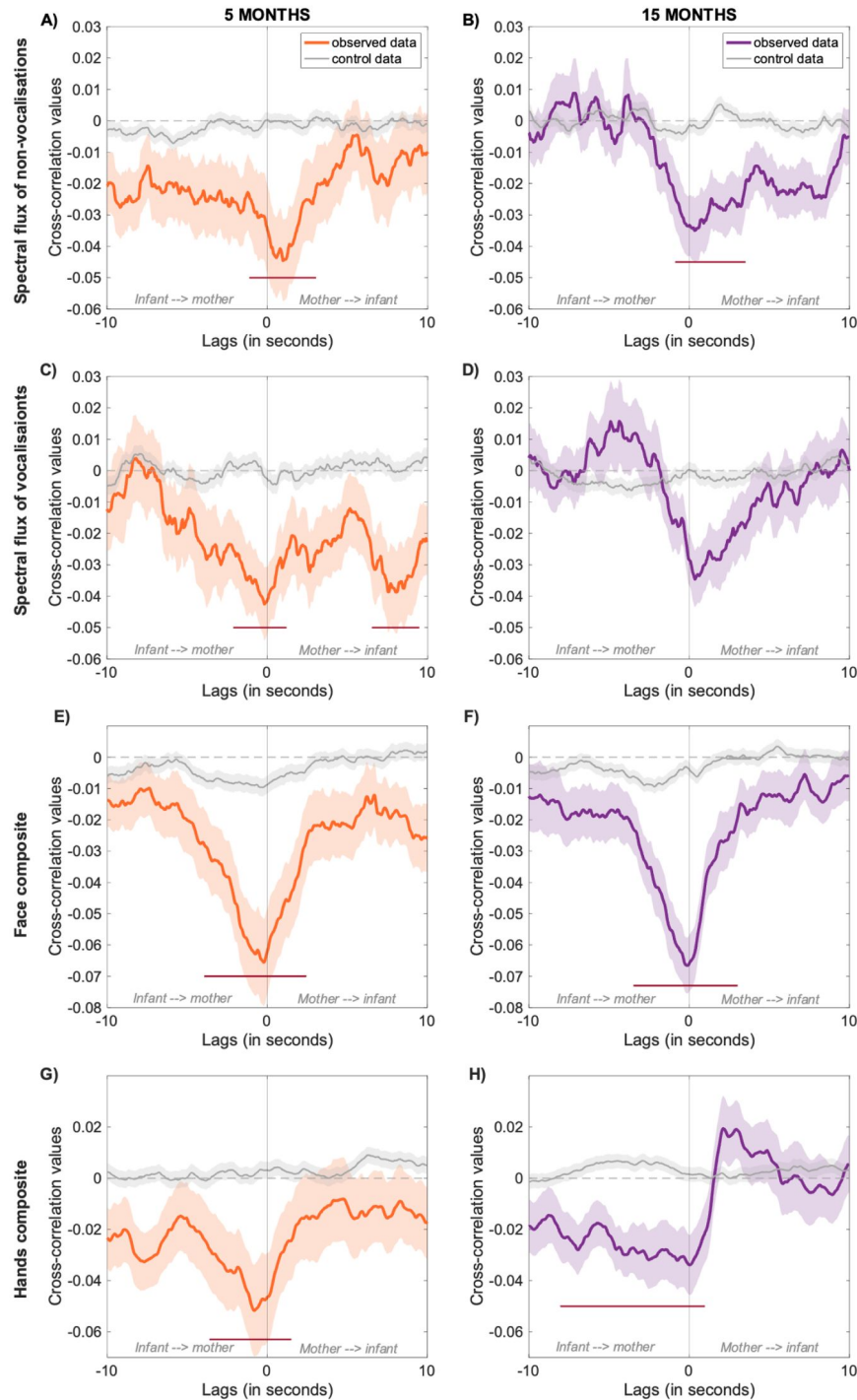


Figure 4.

Cross correlations between lower-level features and infant attention to objects.

Spectral flux of ambient background and vocalisations, facial movement, and hand movement in relation to infant attention to objects at 5 months (A, C, E, G) and 15 months (B, D, F, H), respectively. Thick orange (5 months) and purple (15 months) lines represent the observed cross-correlation results, with shaded coloured areas showing their SEM. Grey lines represent control (permuted) data, with the shaded grey area indicating its SEM. Red thick lines indicate significance from the CBP test (significance for the CBP tests was set to $p < 0.025$, two-sided, and was then FDR adjusted).

3.3.2.3 Semantic surprisal

No significant associations were observed between semantic surprisal and infant attention to objects (**Fig 4E, 4F**) or attention to mothers (**Figure S7E, F**).

3.3.2.4 Facial novelty

We found negative correlations at $t=0$, indicating that increased infant attention to objects is associated with decreased maternal facial variability. These associations were significant at 15 months from $t=-5.68$ (infant precedes mother) to $t=+1.96$ sec (mother precedes infant) ($p<0.001$, $p\text{-adj}=0.003$; **Fig 4H**). At 5 months, instead, these associations were non-significant (**Fig 4G**). Overall, these findings are stronger at backward time-lags indicating that increases in infant attention to objects precede decreases in maternal facial expressions (or that decreases in infant attention to objects precede increases in the variability of maternal facial expressions) (**Fig 4G, H**). As with the face movement data (**Fig 3E, F**), we expected that increases in the variability of maternal facial novelty would associate with infant attention to mothers. However, this was not the case. Although a pattern emerged, where infants appeared to pay more attention to their mothers when they increased the variability of their facial expressions, this was not significant (**Fig SM 7G, H**). The relationship between facial novelty and attention may be related to the relationships already observed between facial salience and attention (**Fig 4E, F**) as the two variables are themselves weakly positively associated ($r=.007/.02$ at 5/15 months).

4. Discussion

In this study we examined mother-infant interactions during dyadic object play at 5 and 15 months. We characterised eight features of the interaction, primarily derived from maternal behaviours, that were further categorized into lower-level features - namely spectral flux of ambient noise and of maternal vocalisations, and maternal face and hand movement - and higher-level features, including information rate, semantic surprisal, toy naming and facial novelty. Using time-series analyses we investigated how these features of the interaction, most of them coming from the behaviours of the mothers, dynamically modulate and are modulated by fine-grained fluctuations in the infant's attention state.

Our preliminary analyses indicated no significant differences in infants' overall attentiveness to the objects between 5 and 15 months (**Fig S2A and B**). Examining interaction features from 5 to 15 months revealed increases in mother object naming, information rate and total amount of speech (**Fig 2E, 2F, S3**), while all other features remained unchanged (**Fig 2**). This is in line with previous research on caregiver speech becoming more complex over infant development (Schwab et al., 2018) in order to accommodate for infant needs for redundancy for word learning (Schwab & Lew-Williams, 2016; Tal et al., 2021).

Next, we examined the relationships among all eight lower- and higher-level interaction features (**Fig 3, S5**). The majority of the features of the interaction showed positive associations, particularly around lag 0, indicating small but significant levels of temporal coordination. This suggests that mothers engage in multiple behaviours across various modalities simultaneously when interacting with their infants, highlighting the interconnected nature of our variables. Interestingly, there were few differences observed between 5 and 15 months, implying that the features of the interaction remain relatively consistent across these time points. Importantly, while higher- and lower-level features were interrelated (**Fig. 3, S5**), they showed distinct patterns in relation to infant attention to objects and to the mothers. This suggests that, despite their interconnectedness, they contribute differently to dyadic interaction dynamics and the shaping of infant attention.

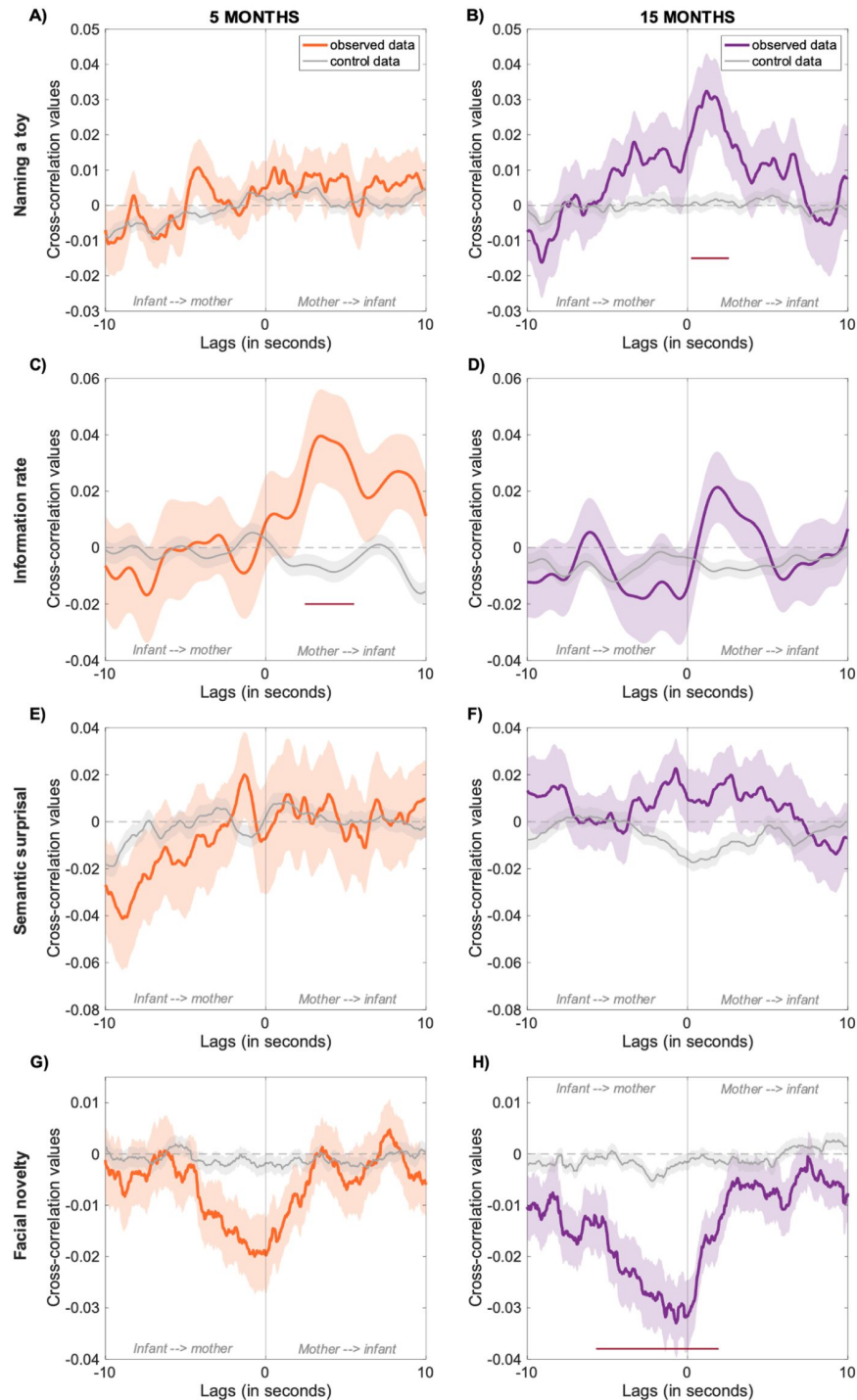


Figure 5.

Cross correlations between higher-level features and infant attention to objects.

Maternal object naming, information rate, semantic surprisal, and facial novelty in relation to infant attention to objects at 5 months (A, C, E, G) and 15 months (B, D, F, H), respectively. Thick orange (5 months) and purple (15 months) lines represent the observed cross-correlation results, with shaded coloured areas showing their SEM. Grey lines represent control (permuted) data, with the shaded grey area indicating its SEM. Red thick lines indicate significance from the CBP test (significance from CBP was set to $p < 0.025$, two-sided; of note, significance for object naming was set at $p < 0.05$, one-sided, based on the expectation that object naming would positively correlate with attention. All results were FDR adjusted).

When we examined the dynamic associations between infants' attention and lower-level features of the mothers' behaviour, we found a range of fine-grained associations, consistent with Prediction 1, which suggests that mothers use multiple behavioural cues during play that are tightly coupled to changes in infant attention (Perapoch Amadó et al., 2025 [↗](#); Phillips et al., 2023 [↗](#); Suarez-Rivera et al., 2019 [↗](#)). We found that ambient spectral flux, spectral flux of vocalisations, face and hand movements all showed negative correlations at time $t=0$, suggesting either that caregivers increased their saliency when infants were not paying attention to the toy objects, or that they decreased their saliency when infants were paying attention, presumably to help them focus on the toys (**Fig 3** [↗](#)). Contrary to Prediction 2, we found that the direction of these associations did not change with age: relationships were generally consistent between 5 months and 15 months. When we introduced a time lag between the variables we found, consistent with Prediction 3, that decreases in spectral flux (a measure of acoustic change) in both maternal vocalisations and the ambient noise was followed by subsequent increases in infants' attention to objects (**Fig 3 A-D** [↗](#)). Additionally, increases in infants' attention to objects were followed by decreases in mothers' hand and facial movements (**Fig 3 E-H** [↗](#)). As expected, maternal facial movements were associated with infant attention to mother's faces (**Fig S6E and S6F** [↗](#)).

Overall, these findings suggest that mothers adapt their low-level behaviour across different modalities to support their infant's attentional focus. Specifically, when infants direct their attention toward an object, mothers may intentionally pause their ongoing actions or interactions to facilitate the infant's engagement with the object of interest. Interestingly, these lower-level features were all positively correlated with infant attention to mothers' face at $t=0$ (the opposite pattern observed with infant attention to objects), suggesting that mothers may reduce these behaviours not only to support infants' attention to objects but also to avoid distracting them from the objects. This aligns with previous research demonstrating that mothers actively scaffold their infants' attention (e.g. Bakeman & Adamson, 1984 [↗](#); Bigelow et al., 2004 [↗](#); Suarez-Rivera et al., 2019 [↗](#); Sun & Yoshida, 2022 [↗](#)). Our findings are interesting because they highlight not only the positive associations between low-level maternal behaviours and infant attention - where increased maternal behaviours correspond to increased infant attention - but also the role of maternal behaviour reduction. More specifically, the decrease in certain maternal behaviours, potentially those that could have been acting as distractors, was also linked to greater infant attention. This suggests an adaptive parental strategy that not only involves increasing certain behaviours to support infant engagement, but also strategically reducing others, highlighting the importance of modulating low-level behaviour contingent on infant attention.

We also observed dynamic associations between infants' attention and higher-level features of the interaction. Consistent with Prediction 1, we found that decreases in infant attention to objects preceded increases in the variability of maternal facial expressions; or, alternatively, that increases in infant attention preceded decreases in the variability of maternal facial expressions (**Fig 4G, H** [↗](#)). We also found that increases in object naming and information rate were positively associated with subsequent increases in infant attention to objects (**Fig 4 A-D** [↗](#)). Object naming predicted attention to objects at 15 months but not at 5 months (**Fig 4A, B** [↗](#)), while information rate showed this relationship at both ages but reached significance only at 5 months (**Fig 4C, D** [↗](#)). These findings are comparable to Suarez and colleagues who found that parental behaviours such as talking and touching were not only highly likely to occur when both the parent and infant visually attended to the same object but were also associated with longer periods of infant attention (Suarez-Rivera et al., 2019 [↗](#)).

Similarly (and perhaps surprisingly, given that information rate and semantic surprisal were the most strongly associated of the behavioural variables (**Fig S5** [↗](#))), we found associations between information rate and infant attention (**Fig 4C, D** [↗](#)) but no associations between semantic surprisal and infant attention (**Fig 4E, F** [↗](#)). This may indicate that 5- and 15-month-old infants lack of sensitivity to higher-order linguistic structures (Hasson et al., 2015 [↗](#); Heilbron et al.,

2022 [↗](#)). Instead, infants at 5 and 15 months might base their predictions on a simpler statistical learning process that tracks transitional probabilities between part of words ([Chater & Vitányi, 2002](#) [↗](#); [Schmidhuber, 2009](#) [↗](#); [Wolff, 2014](#) [↗](#), 2019 [↗](#)). An alternative, and complementary, explanation could be that the “semantic surprisal” variable was calculated using models trained on adult-directed text. Consequently, it is possible that the aspects deemed surprising by the model were not equally surprising to infants. However, the calculations from the model are based on what has been said within the interaction itself, which makes it unlikely that the model’s interpretation of surprisal diverges significantly from what anyone, including infants, could experience in that context. Instead, what these findings might suggest is that even at 15 months, infants are still likely in the process of developing the linguistic and cognitive frameworks necessary to actively infer meaning from language in a very coarse hierarchical scale and are not yet making predictions based on the semantic meaning of words. However, they orient their attention to speech that updates their past representation, potentially revealing that their predictions are based on a lossless compression process that needs updating when facing unseen sequences. In that regard, early attention follows the principle of least effort, where attention is allocated to speech when it offers relevant information for model updating ([Gerken et al., 2011](#) [↗](#)).

Overall, Prediction 2, which proposed that higher-level features would be more influential for older infants, was only partially supported. While associations between object naming and facial novelty with infant attention were stronger at 15 months, associations between information rate and infant attention were stronger at 5 months. Similarly, Prediction 3, which posited that as infants take the lead in play, higher-level maternal behaviours would more often follow infant attention to objects, was also not supported. Even at 15 months, both increased object naming and increased information rate continued to predict subsequent increases in infant attention, rather than following it. This finding is surprising given prior research suggesting that mothers often name the object their infant is already attending to (i.e. object naming/talk follows infant attention more than the other way around) ([Schroer & , 2022](#) [↗](#)), and that this practice facilitates infant learning more effectively than naming objects not currently in the infant’s focus ([Goupil et al., 2024](#) [↗](#); [Yu & Smith, 2012](#) [↗](#)). However, this effect may not necessarily hold if infants already know the words.

Recent research in psychology has challenged traditional theoretical approaches which characterise the physical environment purely in terms of exogenous features, quantified through measures such as salience ([Itti & Baldi, 2009](#) [↗](#)), and attention as the product of a push-pull between exogenous and endogenous factors ([Luna et al., 2008](#) [↗](#)). Rather, we dynamically recalibrate the salience in our environments through how we interact with them - for example, by picking up an object we are interested in and pulling it closer ([Anderson et al., 2022](#) [↗](#); [Franchak & Yu, 2022](#) [↗](#); [Méndez et al., 2021](#) [↗](#); [Schroer & Yu, 2022](#) [↗](#)): we generate experiences through behaviours ([Dewey, 1896](#) [↗](#); [Gibson, 1988](#) [↗](#)). In many ways, the findings from this study extend this idea into social interaction, by showing that our behaviours influence not just the low-level properties of how our partners move and talk, but also what information they present to us, and when. Already during infancy, the information that infants perceive is constantly and dynamically changing over multiple hierarchical levels, contingent on their behaviour and on their attention. Even from early development, attention involves interactive processes which unfold across multiple levels.

In summary, our findings describe how, even during early development, maternal behaviours both depend on, and influence, infants’ attention. This suggests that, even from early life, joint attention involves both lower- and higher-level processes. Interactive processes operate across multiple levels, from salience to semantics.

4.1 Limitations

There are a number of limitations to our study. First, the experimental setup, in which a mother and her infant play together while seated across a table in the lab, cannot fully compare to a real-life interaction (Tamis-LeMonda, 2023 [↗](#)), it also limits the variability of interactional patterns that the dyad can engage in, and might have put pressure on mothers to engage in the interaction more than they would have otherwise (Abney et al., 2020 [↗](#)). However, despite these limitations, we believe the present set-up still preserves important characteristics of early interactions, such as responsivity (Fogel & Garvey, 2007 [↗](#)) and multimodality (Español & Shifres, 2015 [↗](#)). Second, the measures selected for this study provide only a limited representation of the diverse and complex contexts of real-world interactions. For instance, we were unable to capture narrative event structures, which are considered crucial for interpreting everyday settings (Zacks, 2020 [↗](#)). Similarly, higher-order factors, such as play sequences, were beyond the scope of this study but are likely to be closely linked to infant attention, too. Third, the behaviours we quantified interact dynamically, potentially producing effects greater than the sum of their individual contributions (Zhang et al., 2021 [↗](#)); however, we analysed them independently. Although we considered creating composite variables or using Principal Component Analysis (PCA) to work with components, doing so would result in the loss of the “meaning” of the individual variables. Fourth, our sample is homogeneous in terms of ethnicity, culture and socioeconomics and consists only of mothers (Table S1 [↗](#)). It will be useful for future studies to include investigations from more heterogeneous groups/communities (Feldman, 2007 [↗](#); Mundy et al., 2007 [↗](#); Taverna et al., 2024 [↗](#)) as well as to include fathers (Aureli et al., 2022 [↗](#)).

Data sharing statement:

Partial restrictions to the data and/or materials apply. Due to the personally identifiable nature of this data (video and audio recordings from infants and their mothers) the raw data will not be made publicly accessible. Researchers who wish to access the raw data should email the lead author and permission to access the raw data will be granted as long as the applicant can guarantee that certain privacy guidelines can be provided.

Supplementary Materials

Maternal education	5 Months	15 Months
Post-graduate degree	45.45	64.71
Degree	31.82	35.29
F.E. Qualification	4.55	0
A Level	4.55	0
GCSE's	9.09	0
No formal qualification	0	0
Other	4.55	0
Prefer not to answer	0	0

Household income	5 Months	15 Months
Under 16k	5.26	0
16k - 25k	0.00	0
26k - 35k	5.26	6.25
36k - 50k	10.53	6.25
51k - 80k	10.53	18.75
More than 80k	57.89	62.5
Prefer not to answer	10.53	6.25

Maternal ethnicity	5 Months	15 Months
White British	50	56.25
Black British	5	0
Other White	20	18.75
Asian, Indian	10	18.75
Black African	10	0
Mixed – White/Afro-Caribbean	5	0
Latina	0	6.25
Not answered	0	0

Languages spoken at home	5 Months	15 Months
One	65	50
Two	30	43.75
Three	5	6.25

Table S1.

Demographic data at 5 and 15 months.

	5 MONTHS (N)	15 MONTHS (N)
Behaviour coded	33	34
Audio features		
Missing wav file	2	4
Match cannot be found *	1	0
Final N of usable datasets	30	30
Object naming	30	30
Information Rate	30	30
Semantic Surprisal	27	30
Spectral flux – ambient	27	30
Spectral flux – vocalisations	27	30
Video features		
Final N of usable datasets	33	34
Facial Novelty (KL expression)	33	31
Facial movement	32	34
Hand movement	32	34

Table S2.

Table summarising the numbers of datasets included in the analyses for both samples as well as reason for exclusion.

Two-sided t-test	p-value/ p-adj (FDR)	t-stat	df	sd
Information Rate	0.002/ 0.003	-3.189	58	0.013
Facial movement	0.335	-0.972	64	0.004
Hand movement	0.104	-1.650	64	2.126
Spectral flux - ambient	0.278	-1.096	55	82.467
Spectral flux - vocalisations	0.144	-1.481	55	178.657
Semantic surprisal	0.791	0.267	55	1.131

Wilcoxon test	p-value/ p-adj (FDR)	z-value	ranksum
Facial novelty (expression KL)	0.246	-1.159	710
Object naming	0.002/ 0.003	-3.068	707

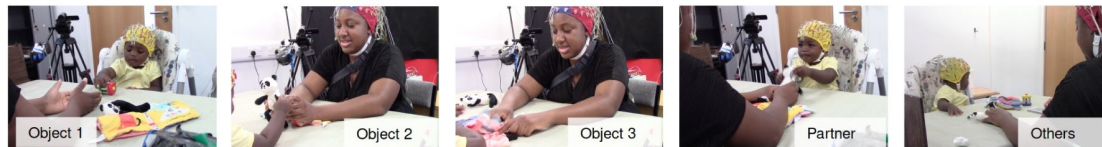
Table S3.

Statistics for overall average levels

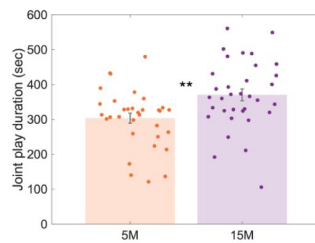
A) Experimental set-up



B) Looking behaviour



C) Average duration (secs) of the interactions



D) Toys employed



Figure S1.

Experimental paradigm

Experimental paradigm. A) Top figure shows the experimental set up for the joint play condition. Two cameras pointed at the infant (view in photos 1 and 2) and one camera pointed at the mother (view in photo number 3). Looking behaviour was coded manually at 50fps for object and partner looks from both the mother and the infant. B) shows the different type of looks (i.e. looks to object 1-3, looks to partner and 'others'). Notice that the latter category - 'others' - included inattention and uncodable moments). C) Plot of the average duration of the joint play interactions at 5 months (in orange) and 15 months (in purple). Asterisks indicate significance from the two-sample t-test (* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$). D) Photos of the toys employed at both time points: panda (A), book (B) and rattle (C).

Figure S2.

Descriptive analysis on looks to the object and to the partner.

Descriptive analyses on looking behaviour. Figure showing average number of looks per minute (A, C) and average look duration (in seconds) (B, D) for looks to object (A, B) and looks to partner (C, D) respectively. Asterisks indicate significance (* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$).

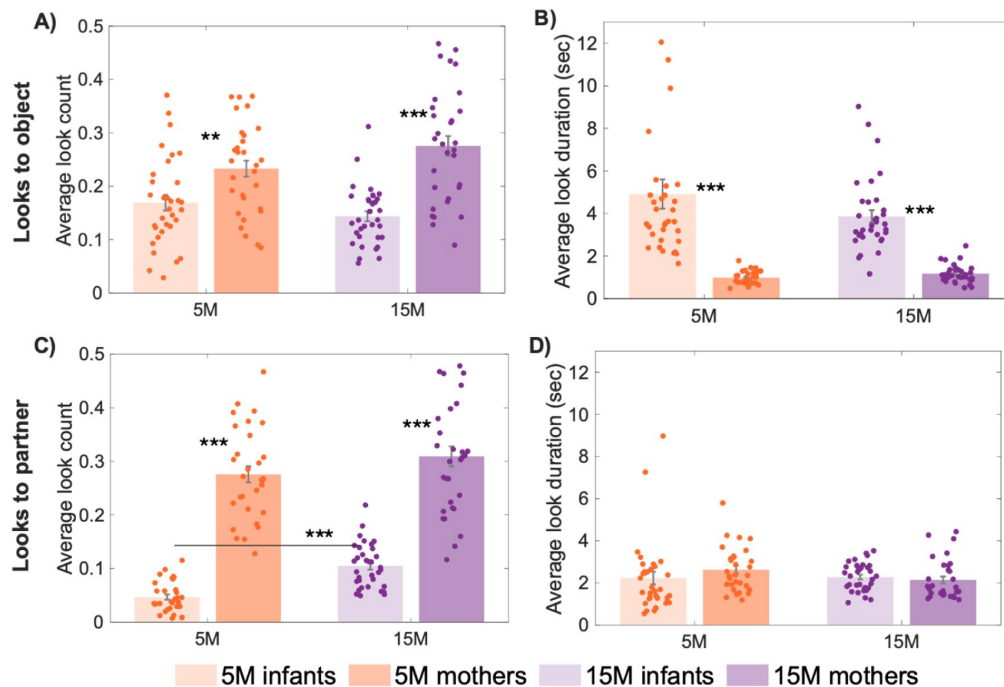
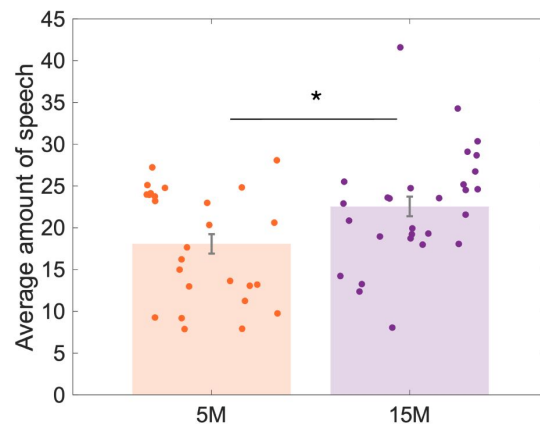


Figure S3.

Average amount of speech per participant.

Descriptive analyses on amount of speech. Figure showing average number of words. To quantify the amount of speech in each interaction, we transformed the output from Whisper Open AI (Radford et al., 2023) into a binary array where ones represent words and zeroes indicate the absence of words. We then summed the number of words per interaction and divided it by the length of each interaction respectively. We conducted a two-sided t-test to assess significant differences across age groups (* indicates $p < 0.05$).



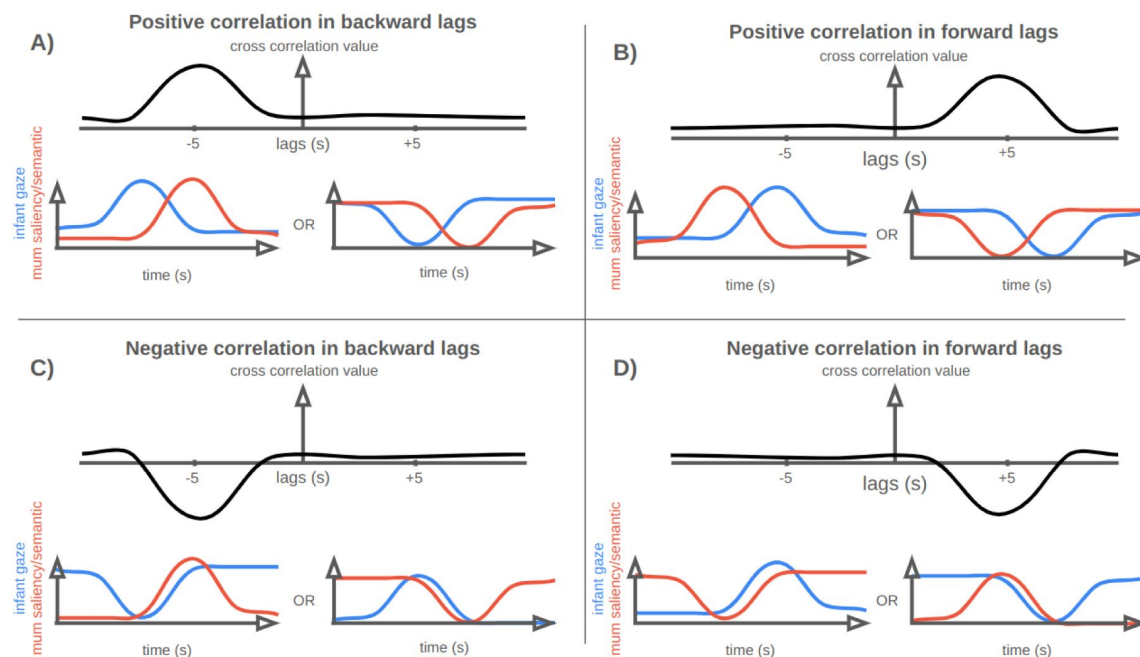


Figure S4.

Schematic illustrating how to interpret cross-correlation results

Visual guide on how to interpret findings from a cross-correlation analysis. Each plot illustrates the interpretation of a positive (A and B) and negative cross-correlation (C and D) values across backward (A and C) and forward (B and D) time-lags. Below each cross-correlation scheme is a illustration of the alternative possible explanations regarding the cross-correlated variables. For instance A) shows a positive correlation in backward lags between infant gaze and maternal behaviour, this can be interpreted is two complementary ways : an increased in infant gaze in followed by an increased in maternal behaviour or that a decreased in infant gaze is followed by a decreased in maternal behaviour.

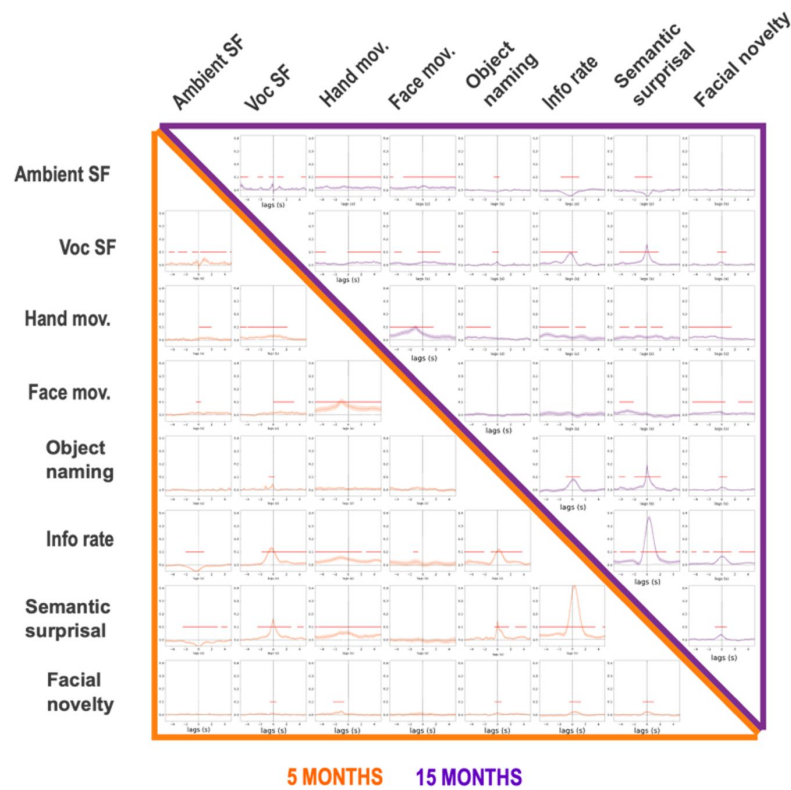


Figure S5.

Correlations between behavioural measures.

Matrix of pairwise cross-correlations between behavioural measures. Each subplot shows the correlation values between pairs of variables at different lags (in seconds), where backwards lags indicate that the first variable (variables at the top) precede the second one (variables on the side), and forward lags that the first variable follows the second one. Please refer to [Fig S4](#) for a visual guide on interpreting findings from

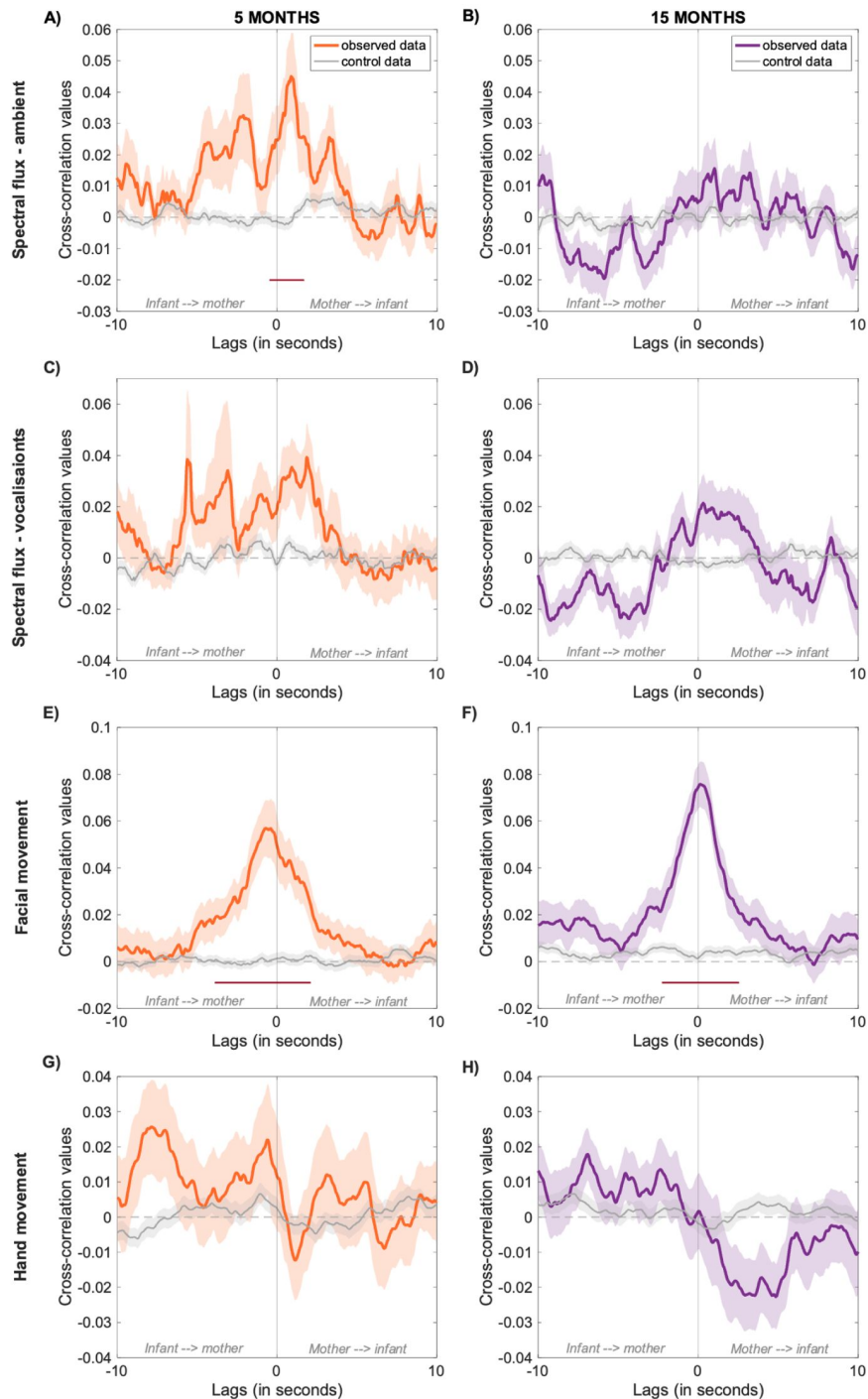


Figure S6.

Cross correlations between lower-level features and infant attention to faces.

Cross correlations between lower-level features and infant attention to faces. Spectral flux of non-vocalisations and vocalisations, facial movement, and hand movement in relation to infant attention to faces at 5 months (A, C, E, G) and 15 months (B, D, F, H), respectively. Thick orange (5 months) and purple (15 months) lines represent the observed cross-correlation results, with shaded coloured areas showing their SEM. Grey lines represent control (permuted) data, with the shaded grey area indicating its SEM. Red thick lines indicate significance from the CBP test ($p < 0.025$, two-sided). All results were FDR adjusted.

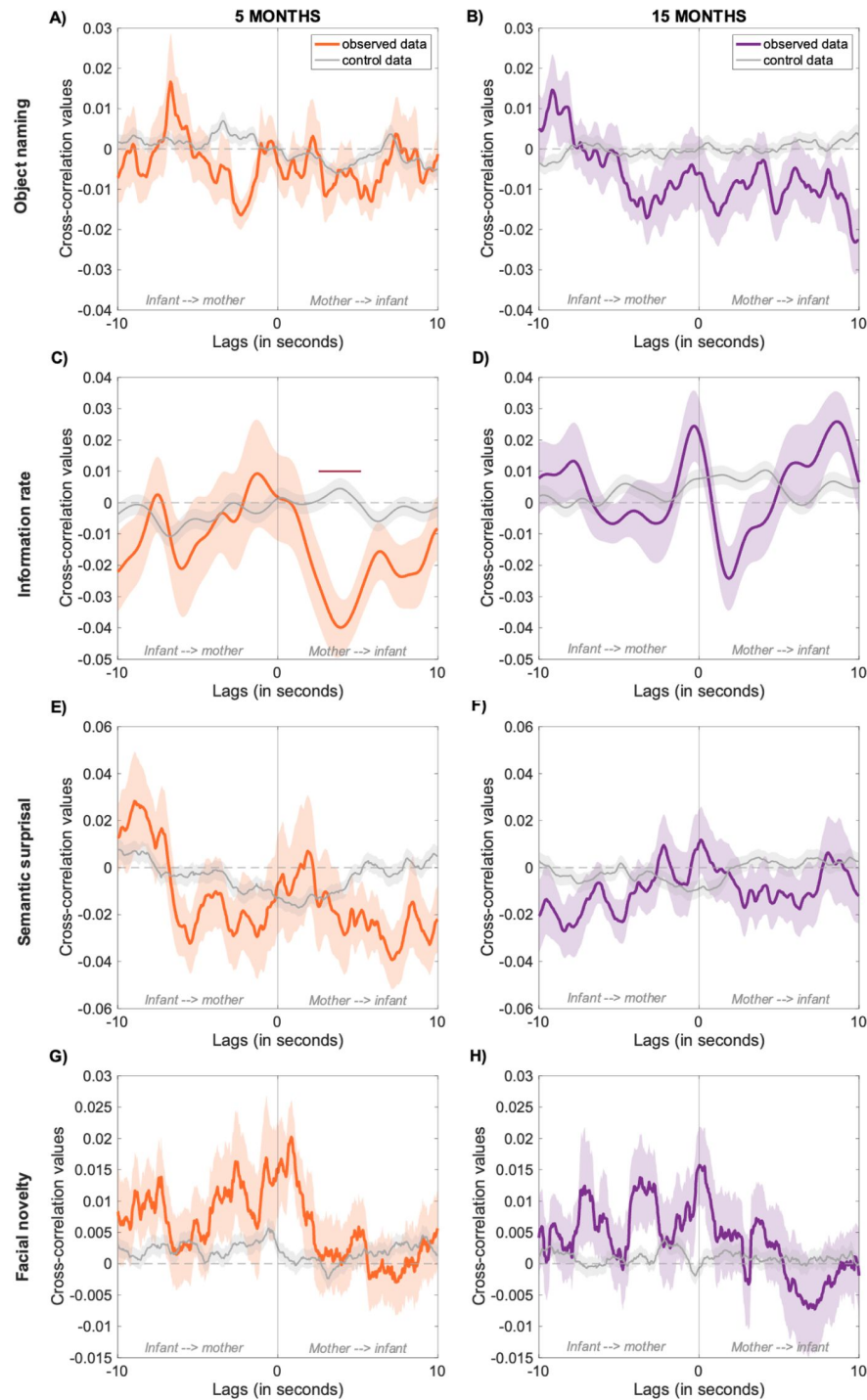


Figure S7.

Cross correlations between higher-level features and infant attention to faces.

Cross correlations between higher-level features and infant attention to faces. Maternal object naming, information rate, semantic surprisal, and facial novelty in relation to infant attention to objects at 5 months (A, C, E, G) and 15 months (B, D, F, H), respectively. Thick orange (5 months) and purple (15 months) lines represent the observed cross-correlation results, with shaded coloured areas showing their SEM. Grey lines represent control (permuted) data, with the shaded grey area indicating its SEM. Red thick lines indicate significance from the CBP test ($p < 0.025$, two-sided). All results were FDR adjusted.

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Reviewer #1 (Public review):

Summary:

This paper investigates infants' social perception as reflected in looking behavior during face-to-face mother-infant toy play in two groups (5 and 15 months). Using information-theoretic and computer-vision methods, the authors quantify dynamic changes in lower-level (salience) and higher-level (semantic) features in the auditory and visual domains - primarily from mothers - and relate these to infants' real-time attention to toys (and to mothers). Time-lagged correlations suggest dynamic, reciprocal relations between infants' attention and maternal low-level (salience) and high-level (semantic) features at both ages, consistent with an early emergence of interpersonal social contingency based on multi-level information during interaction.

Strengths:

The study uses a naturalistic, multimodal mother-infant free-play paradigm and applies information-theoretic/AI methods to quantify both low- and high-level features of maternal behavior, enabling a fine-grained decomposition of interaction dynamics. The time-lag approach further allows examination of temporal relations between maternal signals and infants' attention.

Weaknesses:

Directionality claims from cross-correlations are sometimes unclear, especially when both positive and negative lags are significant, and the evidence for age effects is not yet convincing. Infant attention was manually coded with only moderate-substantial agreement, and handling of disagreements/uncodable periods should be clarified and acknowledged as a limitation.

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Reviewer #2 (Public review):

Summary:

This study examines the dynamic interplay between infant attention and hierarchical maternal behaviors from a social information processing perspective. By employing a comprehensive naturalistic framework, the author quantified interactions across both low-level (sensory) and high-level (semantic) features. With correlation analysis with these features, they found that within social contexts, behaviors such as joint attention - shaped by mutual interaction - exhibit patterns distinct from unilateral responding or mimicry. In contrast to traditional semi-structured behavioral observation and coding, the methods employed in this study were designed to consciously and sensitively capture these dynamic features and relate them temporally. This approach contributes to a more integrated understanding of the developmental principles underlying capacities like joint action and communication.

Strengths:

The manuscript's core strength lies in its innovative, dynamic, and hierarchical framework for investigating early social attention. The findings reveal complex adaptive scaffolding strategies: for instance, when infants focus on objects, mothers reduce low-level sensory input, minimising distractions. Furthermore, the results indicate that, even from early development, maternal behaviors are both driven by and predictive of infant attention, confirming that attention involves complex interactive processes that unfold across multiple levels, from salience to semantics.

From a methodological standpoint, the use of unstructured play situations, combined with multi-channel, high-precision time-series analyses, undoubtedly required substantial effort in both data collection and coding. Compared to the relatively two-dimensional analytical approaches common in prior research, this study's introduction of lower-level and higher-level features to explore the hierarchical organization of processing across development is highly plausible. The psychological processes reflected by these quantified physical features span multiple domains - including emotion, motion, and phonetics - and the high temporal sampling rate ensures fine-grained resolution.

Critically, these features are extracted through a suite of advanced machine learning and computational methods, which automate the extraction of objective metrics from audiovisual data. Consequently, the methodological flow significantly enhances data utilization and offers valuable inspiration for future behavioral coding research aiming for high ecological validity.

Weaknesses:

The conclusion of this paper is generally supported by the data and analysis, but some aspects of data analysis need to be clarified and extended.

(1) A more explicit justification for the selection and theoretical categorization of the eight interaction features may be needed. The paper introduces a distinction between "lower-level" and "higher-level" features but does not clearly articulate the criteria underpinning this classification. While a continuum is acknowledged, the practical division requires a principled rationale. For instance, is the classification based on the temporal scale of the features, the degree of cognitive processing required for their integration, or their proximity to sensory input versus semantic meaning?

(2) The claims regarding age-related differences in Predictions 2 are not fully substantiated by the current analyses. The findings primarily rely on observing that an effect is significant in one age group but not the other (e.g., the association between object naming and attention is significant at 15 months but not at 5 months). However, this pattern alone does not constitute evidence about whether the two age groups differ significantly from each other. The absence of a direct statistical comparison (e.g., an interaction test in a model that includes age as a factor) creates an inferential gap. To robustly support developmental change, formal tests of the Age $\times$ Feature interaction on infant attention are required.

(3) Another potential methodological issue concerns the potential confounding effect of parents' use of the infant's name. The analysis of "object naming" does not clarify whether utterances containing object words (e.g., "panda") were distinct from those that also incorporated the infant's name (e.g., "Look, Sarah, the panda!"). Given that a child's own name is a highly salient social cue known to robustly capture infant attention, its co-occurrence with object labels could potentially inflate or confound the measured effect of object naming itself. It would be important to know whether and how frequently infants' names were called, whether this variable was analyzed separately, and if its effect was statistically disentangled from that of pure object labeling.

(4) Interpretation of results requires clarification regarding the extended temporal lags reported, specifically the negative correlation between maternal vocal spectral flux and infant attention at 6.54 to 9.52 seconds (Figure 4C). The authors interpret this as a forward-prediction, suggesting that a decrease in acoustic variability leads to increased infant attention several seconds later. However, a lag of such duration seems unusually long for a direct, contingent infant response to a specific vocal feature. Is there existing empirical evidence from infant research to support such a prolonged response latency? Alternatively, could this signal suggest a slower, cyclical pattern of the interaction rather than a direct causal link?

Reviewer #3 (Public review):**Summary:**

This manuscript presents an ambitious integration of multiple artificial intelligence technologies to examine social learning in naturalistic mother-infant interactions. The authors aimed to quantify how information flows between mothers and infants across different communicative modalities and timescales, using speech analysis (Whisper), pose detection (MMPose), facial expression recognition, and semantic modeling (GPT-2) in a unified analytical framework. Their goal was to provide unprecedented quantitative precision in measuring behavioral coordination and information transfer patterns during social learning, moving beyond traditional observational coding approaches to examine cross-modal coordination patterns and semantic contingencies in real-time across multiple temporal scales.

Strengths:

The integration of multiple AI tools into a coherent analytical framework represents a genuine methodological breakthrough that advances our capabilities for studying complex social phenomena. The authors successfully analyzed naturalistic interactions at a scale and level of detail that was not previously possible, examining 33 5-month-old and 34 15-month-old dyads across multiple modalities simultaneously. This sophisticated analytical pipeline, combining speech analysis, semantic modeling, pose detection, and facial expression recognition, provides new capabilities for studying social interactions that extend far beyond what traditional observational coding could achieve.

The specific findings about hierarchical information flow patterns across different timescales are particularly valuable and would not have been possible without this sophisticated analytical approach. The discovery that mothers reduce low-level sensory input when infants focus on objects, while increases in object naming and information rate associate with sustained attention, provides new empirical insights into how social learning unfolds in naturalistic settings. The temporal dynamics analyses reveal interesting patterns of behavioral coordination that extend our understanding of how caregivers adaptively modify their responses to support infant attention across multiple communicative channels simultaneously.

The scale of data collection and the comprehensive multi-modal approach are impressive, opening up new possibilities for understanding social learning processes. The methodological innovations demonstrate how modern computational tools can be systematically integrated to reveal new quantitative aspects of well-established developmental phenomena. The computational features developed for this study represent innovative applications of information theory and computer vision to developmental research.

Weaknesses:

Several major limitations affect the reliability and interpretability of the findings. The sample sizes of 33-34 dyads per age group are relatively modest for the complexity of analyses performed, which include eight different features examined across various time lags with extensive statistical comparisons. The study lacks adequate power analysis to demonstrate whether these sample sizes are sufficient to detect meaningful effect sizes, which is particularly concerning given the multiple comparison burden inherent in this type of multi-modal, multi-timescale analysis.

The statistical framework presents several concerns that limit confidence in the findings. Inter-rater reliability for gaze coding shows substantial but not excellent agreement ($\kappa = 0.628$), with only 22% of the data undergoing double coding. Given that gaze coding forms the foundation for all subsequent analyses of joint attention and information flow, this reliability level may systematically influence findings. The multiple comparison correction strategies vary inconsistently across different analyses, with some using FDR correction and others treating lower-level and higher-level features separately. Additionally, object naming analyses employed one-sided tests ($p < 0.05$) while others used two-sided tests ($p < 0.025$) without clear theoretical or methodological justification for these differences.

The validation of AI tools in the specific context of mother-infant interactions is insufficient and represents a critical limitation. The performance characteristics of Whisper with infant-directed speech, the precision of MMPose for detecting facial landmarks in young children, and the accuracy of facial expression recognition tools in infant contexts are not adequately validated for this population. These sophisticated tools may not perform optimally in the specific context of mother-infant interactions, where speech patterns, facial expressions, and body movements may differ substantially from their training data.

The theoretical positioning requires substantial refinement to better acknowledge the extensive existing literature. The authors are working within a well-established theoretical framework that has long recognized social learning as an active, bidirectional process. The joint attention literature, beginning with foundational work by Bruner (1983) and continuing through contemporary theories of social cognition by researchers like Tomasello (1995), has emphasized the communicative and adaptive nature of attentional processes. The scaffolding literature, including seminal work by Wood, Bruner, and Ross (1976), has demonstrated how parents adjust their support based on children's developing competencies. Moreover, there is a substantial body of micro-analytic research that has employed sophisticated quantitative methods to study social interactions, including work by Stern (1985) on microsecond-level interactions and research using time-series methods to examine dyadic coordination patterns.

The cross-correlation analyses have inherent limitations for causal inference that are not adequately acknowledged. The interpretation of temporal correlation patterns in terms of directional influence requires more cautious consideration, as observational data have fundamental constraints for establishing causality. The ecological validity is also questionable due to the laboratory tabletop interaction paradigm and the sample's demographic homogeneity, consisting primarily of white, highly educated, high-income mothers.

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